



Enhancing power capability and fast discharge behavior in P2-type K layered cathodes through structural stabilization via introducing Li-ions into TM layers

Hyunji Kweon^{a,b,1}, Jungmin Kang^{a,b,1}, Bonyoung Ku^{a,b}, Sunha Hwang^{a,b}, Jinho Ahn^{a,b}, Lahyeon Jang^{a,b}, Myungeun Choi^{a,b}, Sang-Yeop Lee^{a,b}, Jihoe Lee^{a,b}, Hoseok Lee^{a,b}, Hun-Gi Jung^{a,c,d}, Jang-Yeon Hwang^{e,f,*}, Hee-Dae Lim^{g,*}, Jongsoon Kim^{a,b,c,d,*}

^a Department of Energy Science, Sungkyunkwan University, Suwon 16419, Republic of Korea

^b SKKU Institute of Energy Science and Technology (SIEST), Sungkyunkwan University, Suwon 16419, Republic of Korea

^c KIST-SKKU Carbon-Neutral Research Center, Sungkyunkwan University, Suwon 16419, Republic of Korea

^d Energy Storage Research Center, Korea Institution of Science and Technology, Seoul 02792, Republic of Korea

^e Department of Energy Engineering, Hanyang University, Seoul 04763, Republic of Korea

^f Department of Battery Engineering, Hanyang University, Seoul 04763, Republic of Korea

^g Department of Chemical Engineering, Hanyang University, Seoul 04763, Republic of Korea

ARTICLE INFO

Article history:

Received 25 April 2025

Revised 6 June 2025

Accepted 9 June 2025

Available online 16 June 2025

Keywords:

K-ion batteries

Cathode

Layered oxide

First-principles calculation

High energy

ABSTRACT

Mn-based layered oxides are widely recognized as cathode materials for potassium-ion batteries (KIBs) due to their high specific capacity derived from their low molar mass. However, the structural instability caused by the Jahn-Teller effect of Mn^{3+} and the large ionic radius of K^+ results in poor electrochemical performance. Herein, we propose an effective structural stabilization strategy for P2-type Mn-based layered oxide cathodes of KIBs through Li-incorporation into the transition metal layer. Using the first-principles calculations and experiments, we demonstrate that the $\text{P2-K}_{0.48}[\text{Li}_{0.1}\text{Mn}_{0.9}]\text{O}_2$ (P2-KLMO) delivers improved electrochemical performance, specific capacity and average discharge voltage of $\sim 124.4 \text{ mA h g}^{-1}$ and $\sim 2.7 \text{ V}$ (vs. K^+/K) at 0.05C ($1\text{C} = 260 \text{ mA g}^{-1}$), outperforming $\text{P2-K}_{0.5}\text{MnO}_2$. *Operando* X-ray diffraction analysis confirms the P2-OP4 phase transition and Mn^{3+} -induced Jahn-Teller distortion are significantly suppressed in P2-KLMO. These improvements are attributed to the lithium introduction into transition metal layers, leading to strengthened structural stability and enhanced K^+ diffusion kinetics. Moreover, synthetic accessibility through the conventional solid-state method provides an additional advantage for practical application of Li-incorporated Mn-based P2-type cathodes in KIBs. We believe our study offers a simple yet effective strategy for designing high-performance and practical cathode materials for KIBs.

© 2025 Science Press and Dalian Institute of Chemical Physics, Chinese Academy of Sciences. Published by Elsevier B.V. and Science Press. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

The reckless use of fossil fuels worldwide leads to environmental pollution problems and serious global warming caused by greenhouse gas emissions [1]. Therefore, the development of eco-friendly energy sources and efficient energy storage systems is

necessary to address these issues [2]. Lithium-ion batteries (LIBs) are receiving significant attention as a promising energy storage system that can be applied not only to portable devices but also to large-scale applications such as electric vehicles (EVs) because of their high energy density and long cycle performance [3,4]. However, due to the limited reserves of lithium sources, it may be challenging to meet the exponentially high demand for LIBs, which leads to an increase in LIB costs. Thus, it is necessary to develop low-cost rechargeable batteries with a stable raw material supply as an alternative to LIBs. Among various candidates, potassium-ion batteries (KIBs) emerged as one of the most

* Corresponding authors.

E-mail addresses: jangyeonhw@hanyang.ac.kr (J.-Y. Hwang), hdlim@hanyang.ac.kr (H.-D. Lim), jongsoonkim@skku.edu (J. Kim).

¹ These authors contributed equally to this work.

promising alternative materials for LIBs due to their abundant potassium resources on earth, low-cost, and alkali ion-based reaction mechanisms similar to LIBs [5]. Notably, KIBs possess a lower redox potential for the K^+/K couple (-2.93 V vs. SHE) than those of Na^+/Na (-2.71 V vs. SHE), and Zn^{2+}/Zn (-0.76 V vs. SHE) [6], suggesting the possibility of achieving higher cell voltages, which may provide an advantage in terms of energy density depending on the system design.

In the case of cathode materials for KIBs, Mn-based layered oxides with prismatic-site-based (P-type) structures were expected to be outstanding materials, owing to their large gravimetric capacities and low production cost [7,8]. However, due to the larger ionic radius of K^+ (~ 1.38 Å) than that of Li^+ (~ 0.76 Å), K^+ de/intercalation accompanies both relatively slow diffusion and large structural change during charge/discharge compared to Li^+ de/intercalation [9]. The large K^+ also induces greater structural strain during intercalation, which limits the extent of reversible redox reactions and reduces the overall accessible capacity [10,11]. Additionally, it was reported that the redox reaction based on Mn^{3+} ions accompanies Jahn-Teller distortion [12]. The Jahn-Teller effect originates from the degenerate e_g orbital of high-spin Mn^{3+} (d^4), which causes axial elongation of Mn–O bonds [13,14]. The accumulation of such local structural distortions during cycling undermines the overall structural integrity of the cathode, ultimately leading to poor electrochemical performance, including reduced power capability and cycling stability.

In this study, we explored the structural stabilization of P2-type Mn-based layered oxide cathodes for KIBs through Li-incorporation into the transition metal (TM) layer. In sodium-ion battery (SIB) systems, previous studies have shown that introducing Li^+ into the TM layer can effectively suppress local lattice distortions and phase transitions, leading to enhanced structural integrity and improved electrochemical performance [15,16]. Building on these insights, we anticipated that this strategy could be effective for P2-type Mn-based layered oxide cathodes for KIBs, which tend to undergo significant structural changes during cycling due to the large ionic radius of K^+ .

P2-type $K_{0.48}[Li_{0.1}Mn_{0.9}]O_2$ (P2-KLMO) delivered enhanced electrochemical performances and structural stability as the promising cathode for KIBs. At the current density of $0.05C$ ($1C = 260$ mA g^{-1}), P2-KLMO exhibited a specific capacity of ~ 124.4 mA h g^{-1} and an average discharge voltage of ~ 2.7 V (vs. K^+/K), whereas the capacity and average discharge voltage of P2-type $K_{0.5}MnO_2$ (P2-KMO) were just ~ 102.7 mA h g^{-1} and ~ 2.4 V (vs. K^+/K), respectively. P2-KLMO also exhibited enhanced power capability and cycling stability compared to P2-KMO.

Operando X-ray diffraction (XRD) and first-principles calculations revealed that introducing Li ions into the TM layers effectively suppressed the P2-OP4 phase transition and Mn^{3+} -induced Jahn-Teller distortion in P2-KLMO compared to P2-KMO, supporting the observed enhancement in electrochemical performance. Furthermore, unlike P2-KMO, which typically requires electrochemical ion-exchange from a Na-based phase (P2- Na_xMnO_2), P2-KLMO can be synthesized directly via high-temperature solid-state reaction. This synthetic accessibility provides an additional advantage for practical application of Li-incorporated Mn-based P2-type cathodes in KIBs.

2. Experimental

2.1. Synthesis of P2- $K_{0.48}[Li_{0.1}Mn_{0.9}]O_2$

P2- $K_{0.48}[Li_{0.1}Mn_{0.9}]O_2$ powder was synthesized using solid-state method with KNO_3 (Alfa Aesar, 99%), $LiNO_3$ (Junsei, 98%), and $MnCO_3$ (Alfa Aesar, 99.9%) as precursors in a molar ratio of

0.48:0.1:0.9. The precursors were sealed in a silicon nitride jar in an Ar-filled glove box to prevent water contamination and mixed using a high-energy ball milling at 400 r min^{-1} for 12 h. The mixture was calcined at 500 °C for 5 h to decompose the nitrates and then calcined again at 850 °C for 12 h (heating rate of 5 °C min^{-1}) in a tube furnace with Air flow (0.3 L min^{-1}). To prevent the formation of byproducts on the surface due to moisture, the calcined powder of P2- $K_{0.48}[Li_{0.1}Mn_{0.9}]O_2$ was promptly moved into an Ar-filled glove box.

2.2. Preparation of P2- $Na_{0.67}MnO_2$ and ion-exchange process for P2- K_xMnO_2

The P2- $Na_{0.67}MnO_2$ powder was synthesized via the solid-state method using Na_2CO_3 (Sigma Aldrich, 99.5%) and MnO_2 (Alfa Aesar, 99.9%) as precursors. They were mixed in the desired ratio and subjected to high-energy ball milling at 400 r min^{-1} for 12 h, with 5 wt % excess of Na_2CO_3 to compensate for losses during the calcination process. The mixture was calcined at 850 °C for 12 h (heating rate of 5 °C min^{-1}) in a tube furnace with Air flow (0.3 L min^{-1}). And then, the P2- K_xMnO_2 electrode was prepared through an electrochemical test based on Na^+/K^+ ion-exchange process, as direct synthesis was difficult. In the ion-exchange process, P2- $Na_{0.67}MnO_2$ electrode was assembled in a K-ion coin cell, and the charge/discharge was performed in the voltage range of 1.5 – 4.1 V (vs. K^+/K) for 100 cycles at $0.5C$ to removal residual Na ions. Finally, the K-ion cell was disassembled in the Ar-filled glove box and washed with dimethyl carbonate (DMC) before being dried.

2.3. Materials characterization

The P2- $K_{0.48}[Li_{0.1}Mn_{0.9}]O_2$ and P2- $K_{0.5}MnO_2$ were analyzed using XRD (PANalytical, Empyrean) with $Mo K_\alpha$ radiation ($\lambda = 0.7093$ Å). The step size was 0.13° and the 2θ range of 4.6° to 34.3° . The patterns were converted to angles with $Cu K_\alpha$ radiation ($\lambda = 1.5417$ Å). Rietveld refinement of the XRD data was performed using FullProf software. The atomic ratios of elements such as K, Li and Mn were analyzed on an inductively coupled plasma-atomic emission spectrometer (ICP-AES) installed at the National Center for Inter-university Research Facilities (NCIRF) at Seoul National University. The valence states of Mn in the structure of P2- $K_{0.48}[Li_{0.1}Mn_{0.9}]O_2$ and P2- $K_{0.5}MnO_2$ were determined from XANES analysis performed on beamline 7D and 10C at the Pohang Accelerator Laboratory (PAL), South Korea. The Mn K-edge spectra were collected in the transmission mode with an electron energy of 2.5 GeV and a current of 200 mA. Mn reference spectra were simultaneously obtained from Mn metal foil. Particle size and distribution measurements were performed using FE-SEM (ZEISS, Gemini SEM 560) and high-resolution TEM (JEM-F200). The *operando* XRD patterns of P2- $K_{0.48}[Li_{0.1}Mn_{0.9}]O_2$ were obtained using an X-ray diffractometer (PANalytical, Empyrean) with $Mo K_\alpha$ radiation $\lambda = 0.7093$ Å in the 2θ range of 5.0° to 24.0° with a step size of 0.01° . An *operando* XRD battery cell (ISBS) made by PANalytical with PDC TECH was used for *operando* XRD experiments. To protect the cell system from water, the beryllium (Be) window was applied.

2.4. Electrochemical characterization

The P2- $K_{0.48}[Li_{0.1}Mn_{0.9}]O_2$ and P2- K_xMnO_2 electrodes were fabricated by the 70 wt% active material, 20 wt% Super P carbon black and 10 wt% polyvinylidene fluoride (PVDF) using *N*-methyl-2-pyrrolidone (NMP) as the solvent. The slurry was applied on Al foil using a doctor blade and dried at 100 °C for 24 h under vacuum conditions. The mass loading of the electrode was ~ 2 mg cm^{-2} . CR2032 cells were prepared using each cathode electrode, K metal

as the counter electrode, a separator (Whatman GF/F glass fiber), and an electrolyte (0.5 M KPF₆ in ethylene carbonate (EC), and diethyl carbonate (DEC) in a volume ratio of 1:1). The coin cells were fabricated in an Ar-filled glove box. Galvanostatic charge/discharge tests were performed at various current rates (0.05C, 0.1C, 0.15C, 0.2C, 0.5C, 1C and 1.5C in the range of 1.5–4.1 V (vs. K⁺/K)) using an automatic battery charge/discharge test system (WBCS 3000, WonATech). The 1C rate was 260 mA g^{−1}, determined based on the theoretical capacity of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂.

2.5. Computational details

All the density functional theory (DFT) calculations were performed using the Vienna *Ab initio* Simulation Package (VASP) [17]. Perdew–Burke–Ernzerhof (PBE) functional of generalized gradient approximation (GGA) was used as an exchange–correlation functional [18], and the projector-augmented wave (PAW) pseudopotentials with a plane-wave basis set [19], which were implemented in the VASP. To consider the electronic localization of Mn and the effect of its 3d orbital on neighboring atoms more specifically at theoretical analysis [20–23], we adopted the GGA + U functional [21] and *U* coefficient of 3.9 eV for Mn [24]. For DFT-D3 calculation [25], which is one of the DFT methods considering the van der Waals force, the structures of K_x[Li_{0.1}Mn_{0.9}]O₂ were set as 2 × 2 × 1 supercells, and *x* was set from 1 to 0 at 0.125 intervals. The configurations in K_x[Li_{0.1}Mn_{0.9}]O₂ were created using the cluster-assisted statistical mechanics (CASM) software [24,26], and the maximum of 20 configurations with the lowest electrostatic energy for each composition were chosen by comparing the Ewald summation. Those selected configurations underwent DFT-D3 calculation and were plotted on the convex-hull plot. The k-point grid was 6 × 6 × 2, a kinetic energy cutoff was fixed at 500 eV, and all the structures were optimized until the force in the unit cell dropped within 0.03 eV Å^{−1} between each calculation step.

The climbing nudged elastic band (CI-NEB) calculations [27–29] were performed to examine the barrier energy that K⁺ ions encountered during their diffusion in K_{0.25}[Li_{0.1}Mn_{0.9}]O₂, which is expected to have the lowest K⁺ content considering the oxidation state of the transition metal. For the CI-NEB calculation, the lattice parameters of the unit cell were fixed while each atom moved. We set two reference K⁺ positions and made five intermediate phases as the diffusion path of the K⁺ ion. The Bader charges of Mn atoms were analyzed up to 3 Å from the nucleus using the VASP, divided into 50 intervals [30].

The theoretical models of K_xMnO₂ were constructed using the same process as K_x[Li_{0.1}Mn_{0.9}]O₂. K_xMnO₂ structures were set as 2 × 2 × 1 supercells, and *x* decreases in 0.25 intervals from 1 to 0. Using CASM software and Ewald summation, a maximum of 20 configurations with the lowest electrostatic energy for each composition were prepared. Through the DFT calculation, the thermodynamically most stable structure for each composition was obtained. All the calculation conditions were identical to those of K_x[Li_{0.1}Mn_{0.9}]O₂.

3. Results and discussion

3.1. Structural and morphological properties of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ (P2-KLMO)

P2-KLMO was successfully prepared using a conventional solid-state method, which is distinct from the other reported P2-type Mn-based layered oxide cathodes prepared by the ion-exchange process based on P2-type Na-based layered oxide [31]. It implies enhanced structural stability in high-temperature heat treatment by Li substitution into the Mn sites. The detailed structure informa-

tion of P2-KLMO was confirmed through Rietveld refinement based on the XRD data. As shown in Fig. S1, it was confirmed that P2-KLMO composed a P2-type layered structure belonging to the P6₃/mmc space group, without impurities or second phases. Moreover, a small peak was observed at ~18.9°, indicating that the introduction of Li⁺ into the TM layers resulted in structural ordering [32]. The detailed structure information of P2-KLMO was confirmed through Rietveld refinement based on the neutron diffraction (ND) data (Fig. 1a). The calculated lattice parameters of P2-KLMO were determined to be *a* (= *b*) = 2.89505(11) Å, *c* = 12.6413(5) Å with low values of reliable factors (*R*_p = 2.78%, *R*_i = 8.91%, *R*_f = 1.95%, and χ^2 = 5.13%), indicating the high reliability of the Rietveld refinement results. Detailed structure information, including the atomic positions, *B*_{iso} values, and occupancies, was provided in Table S1 and Table S3. Moreover, to define the stable Li site in the P2-K_x[Li_{0.1}Mn_{0.9}]O₂, we compared the relative site energy between two theoretical models, one with the Li contained in the octahedral site within the TM layer and another with the Li in the prismatic site within K⁺ layer (Fig. S2). The modelled prismatic site of the Li was the nearest location that could be made through the Li migration from the octahedral site in the TM layer. We drew out the formation energy using the following equation (1). The site energy was derived by subtracting the formation energy of the lithium migrated model from the energy of the model where the lithium is in the transition metal layer (equation (2)).

$$E_{\text{Formation}} [\text{eV/f.u.}] = E_{\text{K}_x[\text{Li}_{0.1}\text{Mn}_{0.9}]\text{O}_2} - ((1-x)E_{\text{K}_0[\text{Li}_{0.1}\text{Mn}_{0.9}]\text{O}_2} + xE_{\text{K}_1[\text{Li}_{0.1}\text{Mn}_{0.9}]\text{O}_2}), \quad (1 \geq x \geq 0) \quad (1)$$

$$E_{\text{Siteenergy}} [\text{eV/f.u.}] = E_{\text{LiinOctahedral site}} - E_{\text{LiinPrismatic site}} \quad (2)$$

The refinement results clearly indicate that Li⁺ ions occupy not the K⁺ sites but the Mn sites in P2-KLMO, which is well consistent with the DFT calculation results. The stabilizing role of Li⁺ in layered oxide cathodes has been reported in several studies, particularly in the Na-ion battery (SIB) system. It is known that Li⁺ incorporation into the TM layer can suppress local lattice distortions and phase transitions, improving structural integrity and electrochemical performance [15,16,33].

Based on the structure information obtained from Rietveld refinement with XRD analyses, the crystal structure of P2-KLMO was illustrated in Fig. 1(b). The P2-KLMO consisted of edge-shared [Li, Mn]O₆ octahedra and prismatic sites of K ions with the ABBA oxygen stacking sequence. P2-type layered structure of P2-KLMO was also confirmed throughout the selected area electron diffraction (SAED) analysis based on the transmission electron microscopy (TEM). P2-KLMO exhibited the presence of the (100), (102), and (104) planes along the [020] direction of a hexagonal phase and confirmed the single-crystalline structure (Fig. 1c). Moreover, the energy-dispersive X-ray spectroscopy (EDS) elemental mapping analyses revealed that the elements of K, Mn, and O were homogeneously distributed in the P2-KLMO (Fig. 1d). The detailed elemental ratio in P2-KLMO was also investigated through the ICP-AES analysis. As tabulated in Table S2, it was verified that the atomic ratio of K: Li: Mn in the P2-KLMO was ~0.48:~0.10:~0.90. The average particle size of ~2 μm was confirmed through scanning electron microscopy (SEM), which was consistent with TEM results (Fig. S3). In addition, XANES analysis was performed to confirm the average oxidation states of Mn elements in the as-prepared P2-KLMO (Fig. 1e). The Mn K-edge XANES spectra of P2-KLMO revealed that the oxidation state of Mn was closer to Mn⁽⁴⁺⁾O₂ reference than Mn₂⁽³⁺⁾O₃ reference, indicating that the average oxidation state of Mn ions in P2-KLMO was expected to be approximately Mn^{3.8+}. We also tried to prepare the P2-K_{0.48}MnO₂ as the control group. However, as shown in Figs. S4 and S5, prepara-

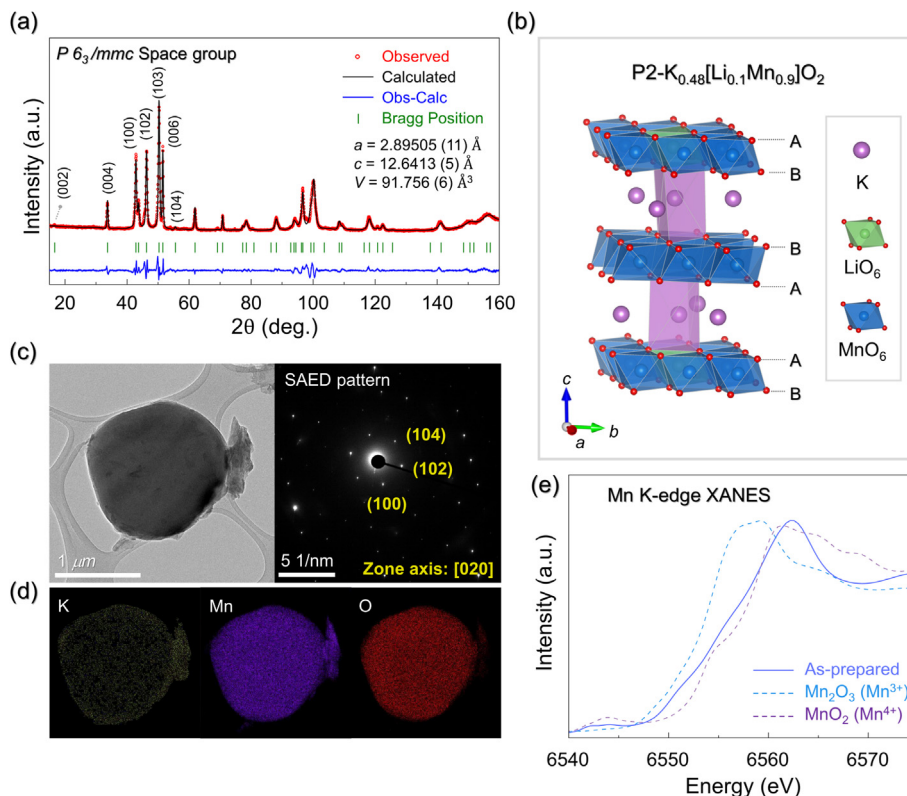


Fig. 1. (a) Refined ND pattern of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂. (b) Crystal structure of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂. (c) TEM image of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and SAED pattern of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂. (d) EDS mappings of K, Mn and O elements. (e) Mn K-edge XANES analysis of as-prepared P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ electrode.

tion of P2-K_{0.48}MnO₂ using the conventional solid-state method resulted in an impure P2-type layered structure and several impurity phases, similar to other reported P2-type layered cathodes for KIBs [34,35]. Thus, we prepared the P2-K_{0.5}MnO₂ (P2-KMO) using the Na⁺/K⁺ electrochemical ion-exchange process from P2-Na_xMnO₂ to P2-K_xMnO₂ (Fig. S6). The detailed process was arranged in the experimental sections. Fig. S7 shows that P2-KMO was successfully prepared as a P2-type layered cathode without any impurities, indicating that the Na⁺/K⁺ electrochemical ion-exchange process was completely achieved. The XRD pattern confirms that P2-KMO possesses a well-defined P2-type layered structure with the P6₃/mmc space group, similar to P2-KLMO. The detailed lattice parameters and volumetric change of as-prepared P2-KMO were provided in Fig. S8 and Table S4. Both P2-NMO and P2-KMO retain a well-defined P2-type layered structure with the P6₃/mmc space group. The replacement of Na⁺ by K⁺ led to the expansion along the c-axis, resulting in an overall increase in unit cell volume. This structural change, attributed to the larger ionic radius of K⁺, is reflected in the shift of the (002) diffraction peak toward lower angles. Importantly, no impurity or secondary phase was detected in the XRD pattern of P2-KMO, and the integrity of the layered structure was well preserved. The SEM images also indicate that the morphology and particle size of P2-KMO are comparable to those of P2-KLMO (Fig. S9). Overall, these results suggest that any structural distortion induced by the ion-exchange process from P2-NMO to P2-KMO had a negligible impact on the electrochemical performance. To investigate the component evolution of any residual during the Na⁺/K⁺ ion-exchange process, ICP-AES analysis was performed. It was revealed that the elemental ratio of Na, K, and Mn in P2-KMO is ~0.00:~0.50:~1.00 (Table S5). In addition, the Li⁺-introduction into the TM layer increases the average Mn oxidation state, resulting in a contraction of the TM-O layers by strengthened Mn–O bonds, and expands the interlayer slab distance [36]. As shown in

Fig. S10, the interlayer slab distance in P2-KLMO was ~4.55 Å, while that of P2-KMO was ~4.39 Å. The enlarged interlayer spacing in P2-KLMO could induce the incorporation of the large K⁺ ions and promote the formation of a well-ordered P2-type layered structure during high-temperature synthesis. This structural stabilization in P2-KLMO can reduce lattice strain and prevents the formation of impurity phases that often occur in P2-KMO under similar conditions.

3.2. Electrochemical performances of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ (P2-KLMO)

To investigate the effectiveness of Li-introduction into the TM layers in enhancing electrochemical performances of P2-type Mn-based layered cathode materials, we compared the power capability of P2-KLMO and P2-KMO. During initial cycling at a current density of 0.05C, P2-KLMO delivered a discharge capacity of ~124.4 mA h g⁻¹ corresponding to ~0.47 mol K⁺ intercalation (Fig. S11a), while P2-KMO just exhibited the capacity of ~102.7 mA h g⁻¹ (Fig. S11b). P2-KMO prepared through an electrochemical ion-exchange process from P2-Na_xMnO₂ via repeated charge/discharge cycling under the KIB system, exhibits similar charge and discharge capacities during the initial cycling. On the other hand, P2-KLMO demonstrates a higher initial discharge capacity than its initial charge capacity due to additional K⁺ intercalation during the initial discharging process, which is similar to the behavior observed in conventional P2-type layered oxide cathodes for SIBs [37–39]. These results also support that P2-KLMO was synthesized via direct synthesis using the conventional solid-state method with high-temperature heat treatment. Moreover, even at a current density of 1.5C, the capacity of P2-KLMO was maintained at ~91.0 mA h g⁻¹ (Fig. 2a and Fig. S12), which is approximately 1.43 times higher than that of P2-KMO under the same conditions (Fig. 2b and Fig. S13). In particular, the difference in electrochem-

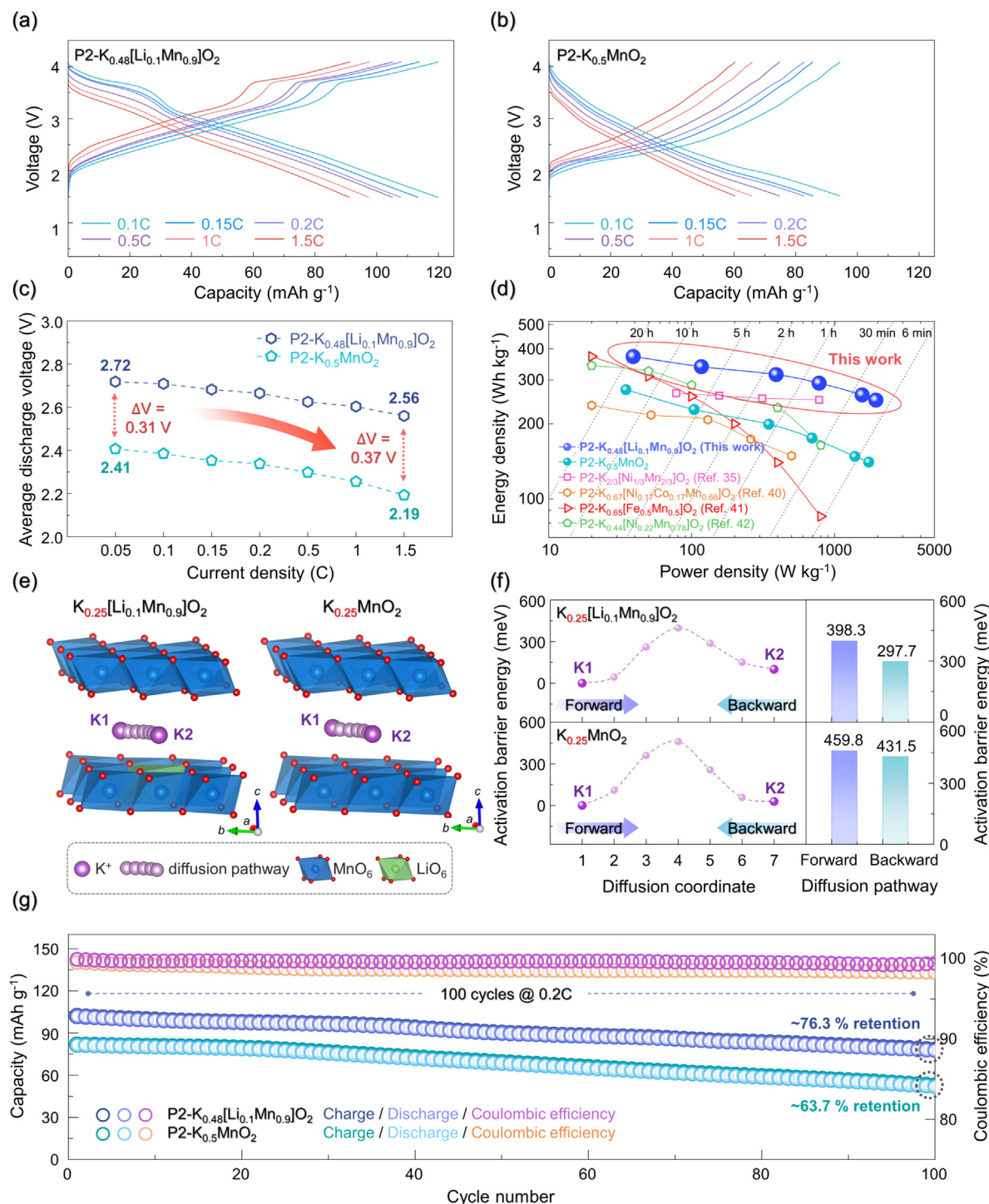


Fig. 2. Charge/discharge profiles of (a) P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and (b) P2-K_{0.5}MnO₂ in the range of 1.5–4.1 V at various current rates. (c) Average discharge voltages of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and P2-K_{0.5}MnO₂ at various current rates. (d) Comparison of energy and power densities among P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and other reported cathode materials for KIBs. (e) Diffusion pathways for K ion with (f) energy landscapes determined using the CI-NEB method in P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and P2-K_{0.5}MnO₂. (g) Cycle performances and Coulombic efficiency of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and P2-K_{0.5}MnO₂ at 0.2C over 100 cycles in the voltage range of 1.5–4.1 V (vs. K⁺/K).

ical properties between P2-KLMO and P2-KMO was also prominent in the comparison of their average discharge voltages. As shown in Fig. 2(c), the voltage gap between P2-KLMO and P2-KMO measured at 0.05C and 1.5C were ~0.31 V and ~0.37 V, respectively. The difference in average discharge voltages increased progressively with the rise in current density. These results indicate that the fast discharge performance and power capability of P2-KLMO are highly enhanced by introducing Li ions into the TM layers. To compare

the electrochemical properties of P2-KLMO with those of other reported cathode materials for KIBs, the Ragone plot was performed (Fig. 2d). The energy densities and power densities of every example, including P2-KLMO and P2-KMO, were calculated using the equations below.

$$\text{Energy density [Wh kg}^{-1}\text{]} = \frac{1}{M_{\text{cell}}} \int_0^t V_{\text{cell}} \cdot idt \quad (3)$$

$$\text{Power density [Wkg}^{-1}] = \frac{1}{M_{\text{cell}}} \int_0^t V_{\text{cell}} \cdot i dt \cdot \frac{1}{t} \quad (4)$$

We organized the P2-type Mn-based KIB cathode materials and their electrochemical performances in a table according to their energy densities in descending order (Table. S6). It was clearly revealed that P2-KLMO delivers high energy density and excellent power capability compared to other layered cathodes for KIBs.

Compared to those for LIBs and SIBs, layered oxide cathodes for KIBs delivered relatively lower capacity due to heavier structural change by the larger ionic radius of K^+ than those of Na^+ and Li^+ . The larger size of K^+ induces greater structural strain during intercalation, which limits the extent of reversible redox reactions and reduces the overall accessible capacity [10,11]. Nevertheless, the P2-KLMO developed in this study achieves relatively high energy density and power capability compared to other K-based layered oxides [35,40–42]. This improvement is attributed to the enhanced structural robustness and suppressed Mn^{3+} -active Jahn-Teller distortion enabled by Li-incorporation, which allows for more efficient K^+ storage and improved electrochemical reversibility. The enhanced power capability and fast discharge performance of P2-KLMO were also confirmed through the climbing NEB (CI-NEB) method based on the first-principles calculation. CI-NEB is known as an effective strategy, which allows a more accurate energy barrier at the hopping points compared to using the simple NEB method [29,43]. As shown in Fig. 2(e, f), the theoretical activation barrier energy for K^+ diffusion in P2-KLMO is ~ 398.3 meV, which is smaller than that in P2-KMO (~ 459.8 meV). The activation barrier energies for K^+ diffusion at other sites were also found to have similar values (Fig. S14). Moreover, except for the bump observed around ~ 3.5 V, P2-KLMO generally exhibited a smaller voltage rise upon constant current application and higher K^+ diffusivity compared to P2-KMO (Fig. S15a). Even though there was a slight decrease in K^+ diffusivity during charging due to the Mn redox reaction, however, K^+ diffusion kinetics recovered rapidly after the bump, and a similar behavior was observed during the discharge process as well. In the case of P2-KMO, a significantly larger overpotential was observed throughout the cycling process compared to P2-KLMO (Fig. S15b). The average ion diffusivity of P2-KLMO was $\sim 1.98 \times 10^{-10} \text{ cm}^2 \text{ s}^{-1}$, whereas that of P2-KMO was $\sim 4.71 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$, clearly indicating a substantial difference in K^+ ion transport kinetics between the two materials. Additionally, from the EIS analysis, the notable enhancement of K^+ ion transfer in P2-KLMO was observed, consistent with the previous electrochemical and computational results (Fig. S16). These results suggest that introducing Li ions into the TM layers facilitates easier K^+ diffusion in P2-KLMO, thereby enhancing power capability in the KIB system. These results suggest that introducing Li ions into the TM layers facilitates easier K^+ diffusion in P2-KLMO, thereby enhancing power capability in the KIB system.

The beneficial effect of Li^+ introduction at the Mn sites is further evidenced by the improved cycling performance of P2-KLMO compared to P2-KMO. As shown in Fig. 2(g) and Fig. S17, the specific capacity of P2-KLMO was retained to $\sim 76.3\%$ of the initial capacity after 100 cycles, while P2-KMO just showed the capacity retention of $\sim 63.7\%$ at the same conditions. Additionally, P2-KLMO exhibited higher capacity retention of $\sim 70.7\%$ compared to approximately 55.9% of retention rate in P2-KMO during 200 cycles at 1C (Fig. S18). The outstanding electrochemical performance of P2-KLMO was further validated through full cell testing. Using P2-KLMO and hard carbon as the cathode and anode, respectively, we conducted full cell tests. Prior to assembly, the hard carbon anode was pre-cycled in the voltage range of 0.01–2.0 V (vs. K^+/K) to mitigate the initial irreversible discharge capacity [44]. The full cell demonstrated excellent cycling stability, retaining $\sim 72.5\%$ of its initial capacity after 100 cycles at 0.2C (Fig. S19).

Additionally, differential capacity versus voltage (dQ/dV) plots for the full cell were analyzed under various cycling conditions (Fig. S20). The results showed that the cathodic/anodic redox peaks of the full cell remained well-maintained after 100 cycles, indicating stable structural properties. These findings suggest that P2-KLMO exhibits excellent electrochemical performances as a promising cathode material for KIBs.

3.3. Structural change of P2- $\text{K}_{0.48}[\text{Li}_{0.1}\text{Mn}_{0.9}]\text{O}_2$ (P2-KLMO) during charge/discharge

We expected that the enhanced electrochemical performances of P2-KLMO would be attributed to the suppressed structural change during Li^+ de/intercalation. Thus, to verify the structural change during charge/discharge we performed the *operando* XRD analyses. In the case of P2-KLMO, several XRD peaks were observed to shift monotonically during the charging process (Fig. 3a, b). Notably, even when charging to 4.1 V (vs. K^+/K), the phase transition from the P2 to OP4 phase was not observed, which is distinctly different from the structural change of P2-KMO that involves the P2-OP4 phase transition, shown in Fig. 3(c, d). We also verified the changes of lattice parameters of P2-KLMO during charge/discharge. As shown in Fig. 3(e), the a - ($=b$) and c -lattice parameters of P2-KLMO changed by only approximately 1.9% and 3.9%, respectively, during charge/discharge. These changes are smaller than those observed in P2-type layered cathodes that undergo the well-known P2-OP4 phase transition. In addition, the (002) diffraction peak of P2-KLMO shifts toward higher angles during the first discharge process, indicating a reduction in interlayer spacing (c -axis contraction). The contraction of the interlayer distance is primarily attributed to the large amount of K^+ ions inserted into the structure during the first discharge. In this initial cycle, the amount of K^+ intercalated during discharge exceeds the amount extracted during the preceding charge, leading to an overall increase in K^+ occupancy within the interlayer sites. This enhanced cation content intensifies the electrostatic attraction between K^+ ions and the adjacent oxygen layers, drawing the transition metal oxide slabs closer together and resulting in a pronounced c -axis contraction. Such electrostatic compression, caused by excess alkali-ion intercalation during the initial discharge, has been commonly reported in P2-type Mn-based layered oxides [38,45,46].

On the other hand, P2-KMO experienced a noticeable contraction of the overall lattice parameters, which was induced by slab gliding and the phase transition from P2 to OP4. Fig. S21 shows the change in lattice parameters of P2- $\text{K}_{0.5}\text{MnO}_2$ structure during K^+ de/intercalation. The *operando* XRD results indicate that the observed differences in the electrochemical profiles between P2-KMO and P2-KLMO are primarily due to the structural stabilization and suppression of local distortions induced by Li-incorporation, rather than a change in the underlying redox chemistry. While both P2-KMO and P2-KLMO operate via the $\text{Mn}^{3+}/\text{Mn}^{4+}$ redox couple, their charge/discharge profiles exhibit distinct features, particularly in the form of multiple sloping regions and differences in voltage distribution (Fig. 2(a, b)). These differences are not attributed to a change in redox mechanism, but rather to structural and electronic stabilization induced by Li incorporation. The DFT-based structural analysis further supports this interpretation. Thus, we suggest that the structural stabilization provided by Li-incorporation can contribute to the electrochemical behavior observed in P2-KLMO. Furthermore, SEM, TEM, and XRD analyses were conducted after extended cycling to examine changes in the crystal structure and morphology of both P2-KLMO and P2-KMO. As shown in Fig. S22, both the crystal structure and morphology were well preserved even after 100 cycles, supporting the stable cycling performance of P2-KLMO in the KIB system. However, after 100 cycles, the P2-KMO particles exhibited severe struc-

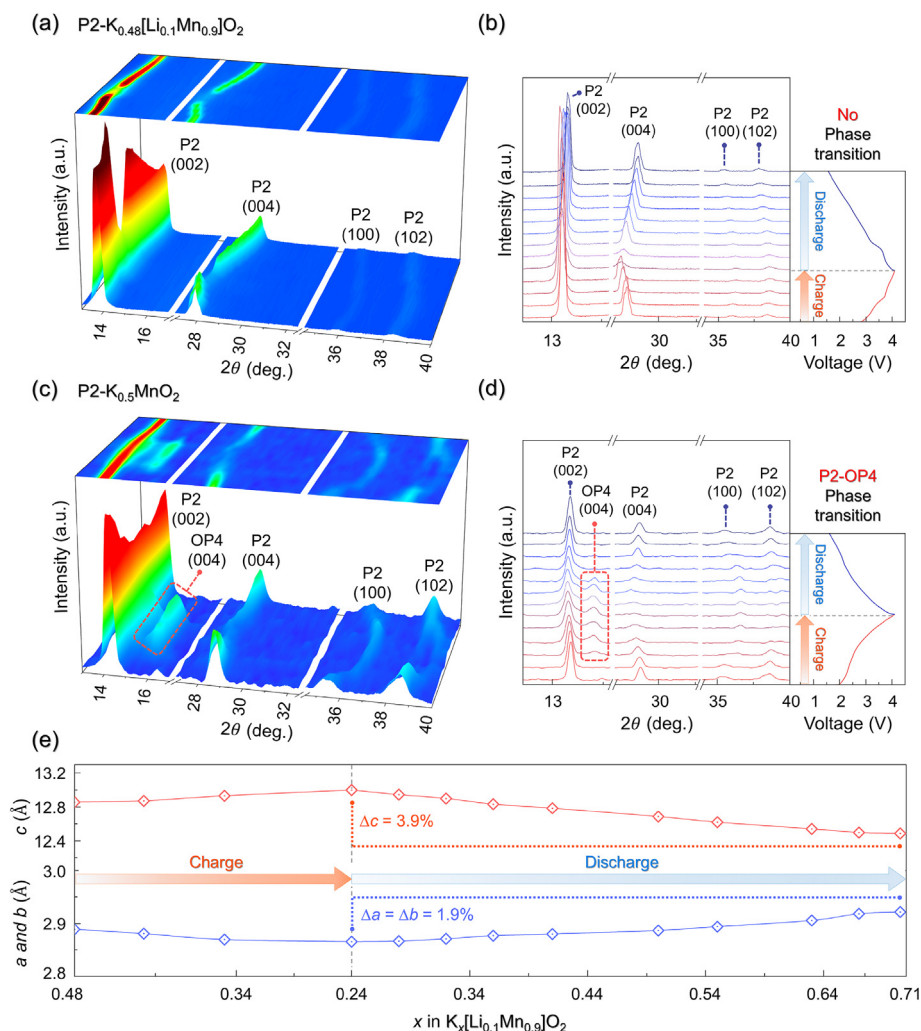


Fig. 3. Operando XRD patterns of (a) P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and (c) P2-K_{0.5}MnO₂ during charge/discharge. The full-operando XRD patterns of (b) P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and (d) P2-K_{0.5}MnO₂ with charge/discharge profile. (e) Change in lattice parameters *a*, *b*, and *c*, which are calculated from the operando XRD results of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ structure during K⁺ de/intercalation.

tural degradation, as evidenced by the formation of prominent cracks (Fig. S23a, b). Moreover, the XRD pattern of P2-KMO showed a significant structural change after cycling (Fig. S23c).

In addition, the suppression of the P2-OP4 phase transition in P2-KLMO was confirmed through first-principles calculations. As shown in Fig. 4(a), K_x[Li_{0.1}Mn_{0.9}]O₂ (K_xLMO) maintains a lower formation energy in the P2 structure than in the OP4 structure, even when the K⁺ content decreases from 0.75 to 0 mol. This suggests that K_xLMO thermodynamically retains the P2 structure during charge/discharge without undergoing the P2-OP4 phase transition. In contrast, for K_xMnO₂ (K_xMO), during K⁺ deintercalation from 0.25 to 0 mol, the OP4 structure becomes thermodynamically more stable than the P2 structure, indicating the occurrence of P2-OP4 phase transition during charge/discharge (Fig. 4(b)). Moreover, the calculation results indicate that the increase in the *c*-lattice parameter was larger in P2-K_xMnO₂ than in P2-K_x[Li_{0.1}Mn_{0.9}]O₂ as the K⁺ content decreased from 0.75 to 0.25 mol, which is consistent with the experimentally observed structural changes in both materials (Fig. S24).

Furthermore, we investigated the changes in the local environment of P2-KLMO and P2-KMO using first-principles calculations. As shown in Fig. 4(c, d), it was confirmed that P2-K_{0.75}MO exhibits a significantly more separated Mn–O bond distance population compared to P2-K_{0.75}LMO. The proportion of Mn–O bonds longer than 2.2 Å was significantly lower in P2-K_{0.75}LMO than in P2-

K_{0.75}MO, indicating that the Mn³⁺-induced Jahn-Teller distortion was effectively suppressed by introducing Li ions into the TM layers. At a K⁺ content of 0.25 mol, the overall Mn–O bond lengths decreased due to Mn oxidation in both materials. In P2-K_{0.25}LMO, the [Mn, Li]–O bond distances were evenly distributed. In contrast, in P2-K_{0.25}MO, a clear separation of Mn–O bond distances persisted, indicating distorted TMO₆ octahedra and associated structural instability. Moreover, the detailed changes on Mn–O bond distance in MnO₆ octahedra are arranged in Fig. 4(e, f). Whereas partial Mn–O bonds are severely changed during K⁺ de/intercalation in P2-KMO, all Mn–O bonds are stably retained in P2-KLMO. Moreover, as depicted in Fig. S25, the Mn atoms exhibited higher Bader charge during the K⁺ de/intercalation, whether it surrounds Li⁺ ion or not. When Li⁺ ions are introduced into the transition metal (TM) layers, they alter the local electronic environment by (i) increasing the average Mn oxidation state (i.e., reducing the fraction of Jahn-Teller active Mn³⁺), and (ii) modifying the local potential landscape, which can lift the *e_g* degeneracy. These electronic-level changes lead to a more symmetric charge distribution, resulting in narrower Mn–O bond length distributions and suppressed structural distortion, as confirmed by several computational approaches.

To experimentally confirm the stabilized local structural environment in P2-KLMO during K⁺ de/intercalation, the

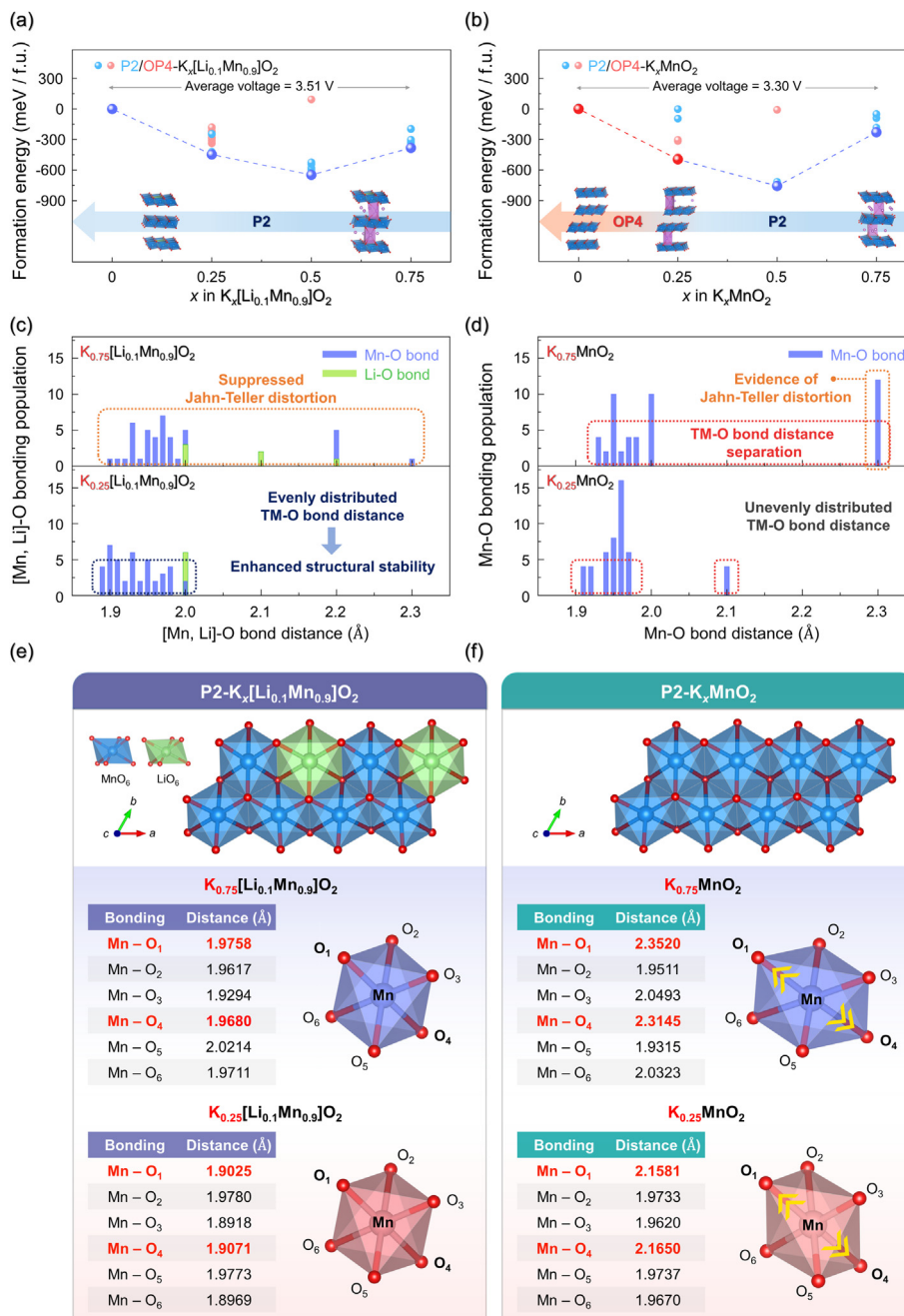


Fig. 4. Convex-hull plots for various (a) $\text{P2-K}_x[\text{Li}_{0.1}\text{Mn}_{0.9}]\text{O}_2$ and (b) $\text{P2-K}_x\text{MnO}_2$ ($x = 0, 0.25, 0.5$, and 0.75) with theoretical average redox potential. [Mn, Li]-O bonding populations of (c) $\text{P2-K}_x[\text{Li}_{0.1}\text{Mn}_{0.9}]\text{O}_2$ and Mn-O bonding populations (d) $\text{P2-K}_x\text{MnO}_2$ ($x = 0.25$, and 0.75). Comparison of the average Mn-O bond distances of MnO_6 octahedra in (e) $\text{P2-K}_{0.75}[\text{Li}_{0.1}\text{Mn}_{0.9}]\text{O}_2$ and $\text{P2-K}_{0.25}[\text{Li}_{0.1}\text{Mn}_{0.9}]\text{O}_2$, and (f) $\text{P2-K}_{0.75}\text{MnO}_2$ and $\text{P2-K}_{0.25}\text{MnO}_2$ determined through first-principles calculation.

synchrotron-based ex situ EXAFS analysis was conducted. As presented in Fig. 5(a, b), the change in Mn-O bond distance during charge/discharge in P2-KLMO was significantly suppressed compared to P2-KMO, indicating that introducing Li ions into the TM layers can enhance the structural stability in P2-KLMO by suppressing Mn^{3+} -induced Jahn-Teller distortion. Furthermore, the EXAFS fitting analysis demonstrated that the change in the average Mn-O bond distance in P2-KLMO during charge/discharge is only $\sim 0.04\%$ (Fig. 5c-f), which is smaller than the $\sim 0.41\%$ change observed in P2-KMO (Fig. S26). These results imply that the introduction of Li ions in Mn sites, which has a fixed oxidation state, significantly affects the stabilization of

the local environment related to the Mn-O bond, which is well matched with the first-principles calculation results. In addition, the synchrotron-based ex situ XANES analysis shows that both P2-KLMO and P2-KMO undergo a $\text{Mn}^{3+}/\text{Mn}^{4+}$ redox reaction during charge/discharge (Fig. 5g, h). After discharge to 1.5 V, the Mn K-edge of P2-KLMO shifts to a lower energy region compared to the as-prepared state. This suggests that additional K^+ intercalation occurs in P2-KLMO during initial discharging process, which aligns with the observation that the discharge capacity is larger than the charge capacity during the initial cycle, which is similar with the reported P2-type Mn-based cathode materials for SIBs [38,46].

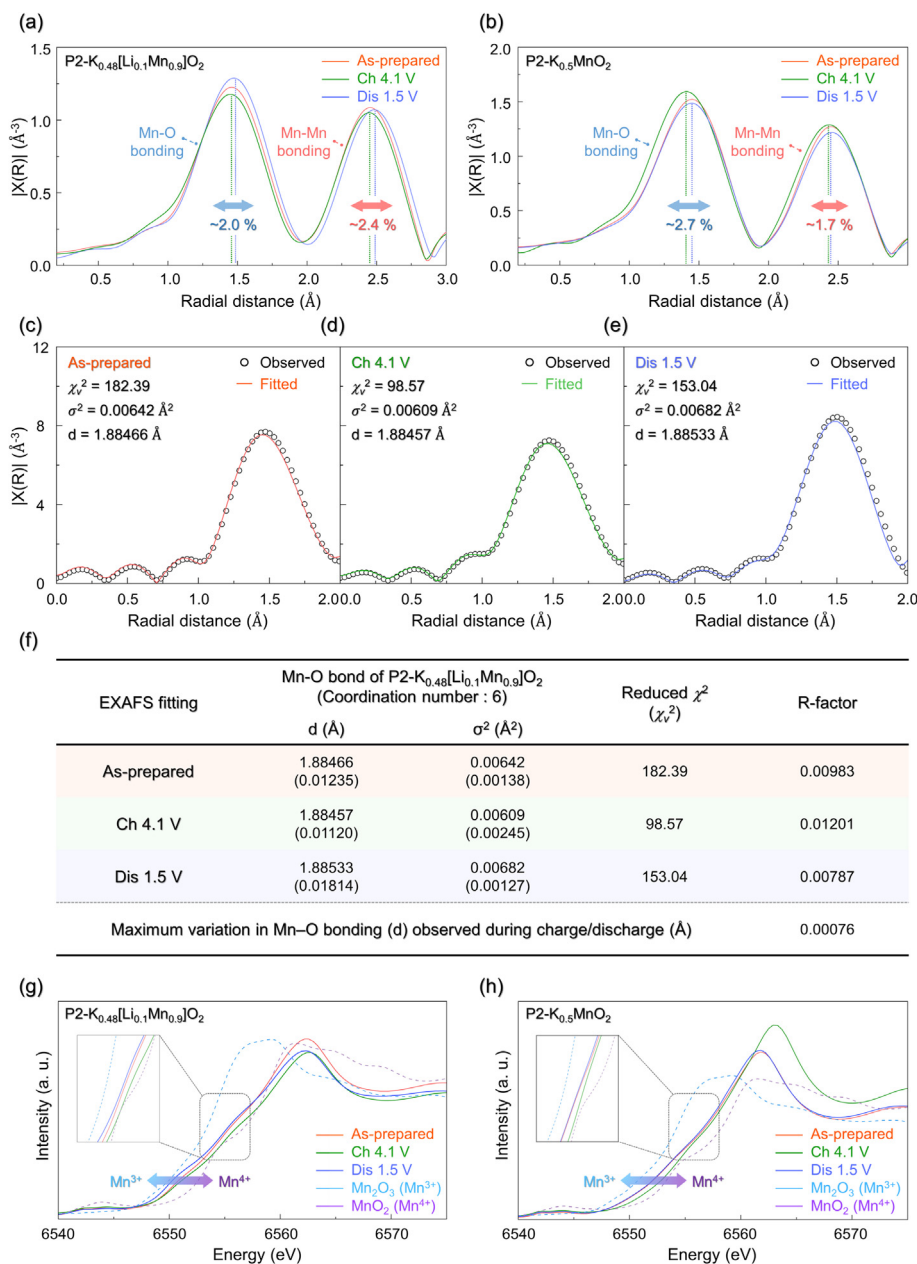


Fig. 5. Ex situ Mn K-edge EXAFS analysis of (a) P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and (b) P2-K_{0.5}MnO₂. Fitted ex situ Mn K-edge EXAFS results of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ at (c) as-prepared, (d) charged state to 4.1 V, and (e) discharged state at 1.5 V. (f) Curve fitting results on ex situ Mn K-edge EXAFS spectra of P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ electrodes at each state and maximum variation of Mn–O bonding observed during charge/discharge. Mn K-edge XANES spectra of (g) P2-K_{0.48}[Li_{0.1}Mn_{0.9}]O₂ and (h) K_{0.5}MnO₂ at as-prepared, full-charge, and full-discharge state.

4. Conclusions

In this study, we demonstrated that Li introduction into the TM layers not only enables the successful preparation of P2-KLMO via a conventional solid-state synthesis method, distinguishing it from other reported layered-type cathodes for KIBs, but also significantly enhances its electrochemical performance and structural stability. XRD and Rietveld refinement analyses confirmed that P2-KLMO, prepared via the solid-state method, exhibited a pure P2-type layered structure without any impurity phases. This is in stark contrast to P2-KMO, which was synthesized through an ion-exchange process rather than a solid-state method. The presence of Li appears to help maintain the layered framework,

enabling access to the target structure without requiring post-synthetic ion exchange. In the electrochemical tests, P2-KLMO exhibited a large specific capacity and high average discharge voltage of $\sim 124.4 \text{ mA h g}^{-1}$ and $\sim 2.7 \text{ V}$ (vs. K⁺/K) at 0.05C, respectively, whereas the capacity and average voltage of P2-KMO were just $\sim 102.7 \text{ mA h g}^{-1}$ and $\sim 2.4 \text{ V}$, respectively. Especially, the voltage gaps between P2-KLMO and P2-KMO were measured to be approximately 0.31 V and 0.37 V at current densities of 0.05C and 1.5C, respectively. Furthermore, the difference of the average discharge voltage between them widens as the current density increases. This demonstrates that Li-substitution at the Mn site greatly improves power capability and fast discharge behavior of P2-KLMO. In the case of cyclability, moreover, the capacity retention

of P2-KLMO for 100 cycles at 0.2C was ~76.3%, which is better than P2-LMO, showing the capacity retention of ~63.7%. The enhanced cyclability of P2-KLMO was supported by detailed structural analyses. In *operando* XRD results, P2-KLMO exhibited reversible structural changes without any phase transitions during charge/discharge cycles. In contrast, P2-KMO clearly undergoes a P2 to OP4 phase transition. Additionally, *operando* XRD and synchrotron-based XAS analysis revealed that the changes in lattice parameters and the local environment during charge/discharge were significantly suppressed in P2-KLMO compared to P2-KMO.

These results clearly demonstrate that the introduction of Li ions into the TM layers enhances the structural stability of P2-KLMO, thereby enabling its improved electrochemical performance. We believe that this discovery not only provides insights into simplifying the synthesis process required for the practical application of high-performance cathode materials for KIBs but also offers a methodology to significantly enhance their electrochemical performance.

CRediT authorship contribution statement

Hyunji Kweon: Writing – original draft, Visualization, Validation, Software, Conceptualization. **Jungmin Kang:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Bonyoung Ku:** Data curation, Formal analysis, Methodology, Visualization. **Sunha Hwang:** Formal analysis, Investigation, Validation, Visualization. **Jinho Ahn:** Validation, Methodology, Investigation. **Lahyeon Jang:** Data curation, Investigation, Methodology. **Myungeun Choi:** Visualization, Validation, Investigation. **Sang-Yeop Lee:** Visualization, Validation, Data curation. **Jihoe Lee:** Visualization, Validation, Investigation. **Hoseok Lee:** Visualization, Validation, Data curation. **Hun-Gi Jung:** Writing – original draft, Formal analysis. **Jang-Yeon Hwang:** Writing – review & editing, Supervision, Methodology. **Hee-Dae Lim:** Writing – review & editing, Validation, Investigation. **Jongsoo Kim:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research was supported by the National Research Foundation of Korea funded by the Ministry of Science and ICT of Korea (RS-2024-00408156).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jechem.2025.06.019>.

References

- [1] S. Xiong, J. Ji, X. Ma, *Waste Manag.* 102 (2020) 579–586.
- [2] J. Kang, J. Ahn, H. Park, W. Ko, Y. Lee, S. Lee, S.K. Jung, J. Kim, *Adv. Funct. Mater.* 32 (2022) 2201816.

- [3] J.H. Yang, Y.K. Jeong, W. Kim, M.A. Lee, J.W. Choi, H. Seung Kim, K.J. Kim, *Adv. Energy Mater.* 15 (2024) 2304366.
- [4] Y. Min Kim, J. In Choi, B. Keun Park, J. Woo Han, J.W. Lee, H. Seung Kim, K. Jae Kim, *Chem. Eng. J.* 486 (2024) 150354.
- [5] W. Ko, S. Lee, H. Park, J. Kang, J. Ahn, Y. Lee, G. Oh, J.K. Yoo, J.Y. Hwang, J. Kim, *Carbon Energy* 6 (2024) e454.
- [6] J.Y. Hwang, J. Kim, T.Y. Yu, S.T. Myung, Y.K. Sun, *Energy Environ. Sci.* 11 (2018) 2821–2827.
- [7] J. Kang, H. Park, W. Ko, Y. Lee, J. Ahn, J.K. Yoo, S.H. Song, H. Kim, J. Kim, *Mater. Chem. A* 9 (2021) 9898–9908.
- [8] X. Wang, Z. Xiao, K. Han, X. Zhang, Z. Liu, C. Yang, J. Meng, M. Li, M. Huang, X. Wei, L. Mai, *Adv. Energy Mater.* 13 (2023) 2202861.
- [9] J.U. Choi, J. Kim, J.H. Jo, H.J. Kim, Y.H. Jung, D.C. Ahn, Y.K. Sun, S.T. Myung, *Energy Storage Mater.* 25 (2020) 714–723.
- [10] Y. Zheng, H. Xie, J. Li, K.S. Hui, Z. Yu, H. Xu, D.A. Dinh, Z. Ye, C. Zha, K.N. Hui, *Adv. Energy Mater.* 14 (2024) 240461.
- [11] N. Yabuuchi, M. Kajiyama, J. Iwatate, H. Nishikawa, S. Hitomi, R. Okuyama, R. Usui, Y. Yamada, S. Komaba, *Nat. Mater.* 11 (2012) 512–517.
- [12] F. Li, X. Gu, S. Wu, S. Dong, J. Wang, P. Dai, L. Li, D. Liu, M. Wu, *Electrochim. Acta* 439 (2023) 141571.
- [13] P.-F. Wang, T. Jin, J. Zhang, Q.-C. Wang, X. Ji, C. Cui, N. Piao, S. Liu, J. Xu, X.-Q. Wang, C. Wang, *Nano Energy* 77 (2020) 105167.
- [14] M.A. Halcrow, *Chem. Soc. Rev.* 42 (2013) 1784–1795.
- [15] Y. Tang, H. Liu, X. Zhu, M. Liu, W. Tang, G. Wei, Y. Feng, B. Xiao, P. Wang, *ACS Nano* 19 (2025) 21118–21129.
- [16] Y. You, S. Xin, H.Y. Asl, W. Li, P.-F. Wang, Y.-G. Guo, A. Manthiram, *Chem* 4 (2018) 2124–2139.
- [17] G. Kresse, J. Furthmüller, *Comput. Mater. Sci* 6 (1996) 15–50.
- [18] J.P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* 77 (1996) 3865–3868.
- [19] P.E. Blöchl, *Phys. Rev. B* 50 (1994) 17953–17979.
- [20] L. Wang, T. Maxisch, G. Ceder, *Phys. Rev. B* 73 (2006) 195107.
- [21] C. Li, P. Li, L. Li, D. Wang, X. Gao, X.J. Gao, *RSC Adv.* 11 (2021) (1856) 21851–21852.
- [22] S.P. Ong, W.D. Richards, A. Jain, G. Hautier, M. Kocher, S. Cholia, D. Gunter, V.L. Chevrier, K.A. Persson, G. Ceder, *Comput. Mater. Sci* 68 (2013) 314–319.
- [23] F. Zhou, M. Cococcioni, C.A. Marianetti, D. Morgan, G. Ceder, *Phys. Rev. B* 70 (2004) 235121.
- [24] A. Jain, G. Hautier, S.P. Ong, C.J. Moore, C.C. Fischer, K.A. Persson, G. Ceder, *Phys. Rev. B – Condens. Matter Mater. Phys.* 84 (2011) 045115.
- [25] S. Grimme, J. Antony, S. Ehrlich, H. Krieg, *J. Chem. Phys.* 132 (2010) 154104.
- [26] A. Van der Ven, J.C. Thomas, Q. Xu, J. Bhattacharya, *Math. Comput. Simul* 80 (2010) 1393–1410.
- [27] G. Henkelman, H. Jónsson, *J. Chem. Phys.* 113 (2000) 9978–9985.
- [28] G. Henkelman, B.P. Uberuaga, H. Jónsson, *J. Chem. Phys.* 113 (2000) 9901–9904.
- [29] P. Maragakis, S.A. Andreev, Y. Brumer, D.R. Reichman, E. Kaxiras, *J. Chem. Phys.* 117 (2002) 4651–4658.
- [30] G. Henkelman, A. Arnaldsson, H. Jónsson, *Comput. Mater. Sci* 36 (2006) 354–360.
- [31] P.K. Jha, A. Golubnichiy, D. Sachdeva, A. Banerjee, G. Sai Gautam, M. Fichtner, A. M. Abakumov, P. Barpanda, *Adv. Funct. Mater.* 34 (2024) 2410665.
- [32] H. Zhou, Y. Bai, C. Yang, C. Guo, F. Liu, P. Hu, C. Han, X. Wang, *Chem. Eng. J.* 488 (2024) 150809.
- [33] M. Choi, H. Ahn, H. Park, Y. Lee, J. Ahn, B. Ku, J. Kim, W. Ko, J. Kang, J.-K. Yoo, D. Kim, J. Kim, *J. Energy Chem.* 85 (2023) 144–153.
- [34] J.H. Jo, J.U. Choi, Y.J. Park, Y.H. Jung, D. Ahn, T.Y. Jeon, H. Kim, J. Kim, S.T. Myung, *Adv. Energy Mater.* 10 (2020) 1903605.
- [35] M.G.T. Nathan, N. Naveen, W.B. Park, K.-S. Sohn, M. Pyo, *J. Power Sources* 438 (2019) 226992.
- [36] C. Liu, S. Luo, H. Huang, Y. Zhai, Z. Wang, *ChemElectroChem* 6 (2019) 2308–2315.
- [37] M. Zarrabeitia, I. Salazar, B. Acebedo, M.Á. Muñoz-Márquez, *Commun. Mater.* 5 (2024) 130.
- [38] J. Hwang, J. Kim, T. Yu, Y. Sun, *Adv. Energy Mater.* 9 (2019) 1–10.
- [39] S.H. Jeong, I.K. Kim, S. Eom, H. Hwang, Y.H. Jung, J.H. Kim, *Energy Storage Mater.* 75 (2025) 104041.
- [40] C. Liu, S. Luo, H. Huang, Z. Wang, A. Hao, Y. Zhai, Z. Wang, *Electrochem. Commun.* 82 (2017) 150–154.
- [41] T. Deng, X. Fan, J. Chen, L. Chen, C. Luo, X. Zhou, J. Yang, S. Zheng, C. Wang, *Adv. Funct. Mater.* 28 (2018) 1–9.
- [42] X. Zhang, Y. Yang, X. Qu, Z. Wei, G. Sun, K. Zheng, H. Yu, F. Du, *Adv. Funct. Mater.* 29 (2019) 1–9.
- [43] J. Lin, K. Jun, G. Ceder, *J. Mater. Chem. A* 2 (2025) 15659–15672.
- [44] H. Park, W. Ko, Y. Lee, J. Kang, J. Ahn, J.K. Yoo, J. Kim, *J. Mater. Chem. A* 9 (2021) (1811) 11802–11801.
- [45] E. Talaie, V. Duffort, H.L. Smith, B. Fultz, L.F. Nazar, *Energy Environ. Sci.* 8 (2015) 2512–2523.
- [46] S. Lee, W. Ko, H. Park, Y. Lee, J. Kang, J. Ahn, S. Lee, E. Sim, K. Ihm, K.-Y. Park, J. Kim, *Chem. Eng. J.* 451 (2023) 138883.